WRYM: The Worm Arena

An open, deterministic game benchmark for language models, with certified solvability, an evidence-based difficulty index, and community level governance

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Abstract

Grid puzzle games are a uniquely clean instrument for measuring planning in large language models: state is fully observable and compactly serialisable, rules are deterministic, and difficulty is structural rather than perceptual. WRYM (a deliberate misspelling of its ancestor WOIM) turns a grid-worm puzzle — eat every apple, manage your own body, push crates, press plates, respect one-way valves, cross crumbling bridges, slide on ice, cut yourself in half when flesh is the only weight that holds a door open, and carry gravity-bound eggs whose support you must never destroy — into a fully static, fully reproducible LLM benchmark. Models see the board only as text and must reason turn by turn under a tight budget. Every level is certified solvable by an exhaustive breadth-first solver before entering the ladder; every recorded run is re-played and signed by a deterministic checker; difficulty is measured by an eight-dimension index computed from evidence on the optimal line. The arena ships a browser level editor, a GitHub-bot review pipeline with maintainer approval for community levels, plug-in adapters for fourteen provider kinds plus Model Context Protocol (MCP), and a leaderboard whose every number is a pure function of committed JSON. The ladder tops out at The Rubicon and The Hatchery, certified 8.4 and 8.0 (ABYSS) under the v4 index — engineered from primary research on Snakebird, Stephen's Sausage Roll, Sokoban's PSPACE-completeness proofs, and WOIM's own carrying-eggs mechanic.

TL;DR

We make LLMs play a hard worm puzzle and measure them honestly. Text-only board, turn budget, exhaustive-solver par, deterministic checker, eight-dimension difficulty index that refuses to inflate, pooled Bradley-Terry ratings with bootstrap confidence intervals, fine-grained progress scores, capability profiles, certified mirror variants, community levels through a bot-and-maintainer review framework, any model pluggable (OpenAI-compatible adapters or MCP), everything committed as JSON and rendered by a static site. Failure is expected — the information is in the failure.

Keywords: language models, benchmarks, game arenas, planning, deterministic evaluation, PSPACE-completeness, Elo rating, reproducibility, Model Context Protocol

1 Introduction

Static question banks saturate and contaminate: answers leak into pretraining corpora, and a percentage on such a bank says increasingly little about planning. Live arenas solved the contamination half by generating fresh comparisons — Chatbot Arena [1] ranks models by pairwise human preference and has become one of the most referenced leaderboards in the field — but preference voting measures conversational polish, not multi-step planning under hard constraints. WRYM occupies the missing quadrant: a live, generative task where the *work* is planning, the rules are fixed, and every artifact of evaluation is public.

The design brief was taken from a commercial game, not from ML literature. WOIM [8] (Dogspawn, 2026) is a Sokoban-tagged worm puzzle on Steam whose core mechanic is that the player's own body — grown by eating apples,

shortened by self-amputation — is simultaneously the mover and the tool. We rebuilt a text-projection dialect of that game with a 19-character grid vocabulary, made an exhaustive solver the arbiter of truth, and wrapped the result in the governance machinery of a serious open-source project: determinism, certification, review, and archival. This paper documents the benchmark (Sections 3-6), its community governance framework (Section 7), its model plug-in architecture including verified free-tier entry points (Section 8), and the current archive (Section 9).

2 Related Work and Design Lineage

LLM arenas. Chatbot Arena [1] established the pairwise-comparison arena as an evaluation instrument: anonymous side-by-side battles, ratings from votes, open data. Its statistical machinery converged on pooled Bradley-Terry maximum-likelihood estimation with bootstrapped confidence intervals. WRYM borrows the arena *shape* and the estimator — all-pairs battles on shared levels fitted at once, with a solved run beating an unsolved one and efficiency breaking ties — but replaces human preference with a deterministic checker, so the ground truth needs no trust in judges. Kaggle Game Arena [11] applies the same doctrine to competitive games; we adopt its head-to-head matrix and its evergreen answer — a new map *is* a new test. Independent third-party measurement sites such as Artificial Analysis [7] play the complementary role of context: our leaderboard links every ranked model to its Artificial Analysis page so arena results can be read against independent intelligence, speed and price measurements.

Game benchmarks for LLMs. Five systems directly shaped the v2 measurement stack. BALROG [12] aggregates six long-horizon game environments and showed both that fine-grained *progress* metrics beat binary win/loss and that frontier models often perform worse with pixels than with text — we adopted the progress score (apples, goal, distance closure on every archived run) and the text-first doctrine. SmartPlay [13] decomposes agent skill into nine named capabilities and maps each game to the subset it stresses; our capability taxonomy (spatial, planning, dependencies, state-tracking, long-horizon, risk, precision) is derived deterministically from each level's geometry and every model carries an intensity-weighted capability profile. LMGame-Bench [14] evaluates LLMs on real games with and without a gaming harness and hardens them against contamination; our answer to contamination is the certified mirror-variant registry — every ladder level has a validator-certified vertical-mirror twin with identical par, so memorization without reasoning is detectable. VideoGameBench [15] demonstrated the modality axis (raw-pixels VLM play); we record modality as an explicit run-schema axis while shipping the deterministic text projection as primary. Terminal-Bench [17] publishes resolution rates with confidence intervals, cost and token columns and structured error analysis; our leaderboard adopted all three, plus its failure taxonomy. AgentBench [16] established the environment-as-pluggable-protocol doctrine our engine, checker and rating layers follow.

Complexity theory. The game family WRYM belongs to is provably hard. Sokoban is PSPACE-complete [2]; the Hearn-Demaine nondeterministic constraint logic framework [3] unified sliding-block and pushing puzzles under one PSPACE-complete umbrella; and food-collecting Snake — the closest formal cousin of our worm — is PSPACE-complete on general grid graphs [4]. Practical consequence: no polynomial shortcut will certify our levels, so the exhaustive solver is not an implementation convenience but the only available ground truth, and board sizes must stay small enough for exhaustive search to terminate inside CI (our ceiling is 25 million states).

Game design lineage. The ladder's difficulty vocabulary was reverse-engineered from the games that define the genre's upper bound: Snakebird [9] (unwinnable-from-one-wrong-press planning, fruit-order length contracts), Stephen's Sausage Roll [10] (long chains of irreversible acts where undo is impossible), and WOIM [8] itself (the body-as-resource economy). Section 6 details how Tier V was engineered from these sources.

3 The Game

A level is a rectangular grid over a 19-character vocabulary (Table 1). The worm is a directed segment chain: a head @ and a body of o cells that trails it. Each turn the model emits exactly one action — up, down, left, right, or split. Movement rules are the classic worm contract with WOIM's amendment: eating an apple grows the worm by one segment; splitting severs the tail beyond the midpoint, permanently donating it as inert terrain. Gates open only while

every floor button is weighted; padlocks consume keys; soft dirt is dug through and never returns; bridges crumble one turn after their last weight leaves; ice slides the worm until something stops it; spikes kill instantly, including mid-slide; one-way valves permit entry only along their arrow; and eggs are gravity-bound cargo — the cell directly beneath an egg must remain solid (terrain, crate, remains, or any worm segment), or the egg falls, cracks, and the run is lost, even on the winning move. The run is won when the last apple has been eaten and the head occupies the heart X, and lost on death or turn-budget exhaustion.

char	element	behaviour
#	soil wall	impassable bedrock; the board frame
.	tunnel	carved open floor
@	worm head	exactly one; directed
o	worm body	trail segments; length is state
*	apple	each eaten apple grows the worm by one
X	heart goal	win tile; unsealed by the last apple
$	walnut crate	pushable, weighs buttons, floats on nothing
%	floor button	depressed by any weight
D	golden gate	open while every button is pressed
~	spikes	instant death, also mid-slide
+	soft dirt	diggable; permanently opens
k / L	key / padlock	crawl over to pick up; consumes one key
> < ^ v	one-way valves	enter only along the arrow
=	crumble bridge	crosses once, then rubble
_	frictionless ice	slide until stopped
O	egg	gravity-bound: support below or it cracks (loss)

Table 1 — The 20-character grid vocabulary (two pairs share rows for space).

4 The Arena Protocol

The pipeline is a fixed, auditable chain: GENERATE, VALIDATE, SOLVE, PLAY, CHECK, ELO, COMMIT, PAGES. **Validate:** the exhaustive breadth-first solver proves each level solvable and computes the optimal move count (par) and the difficulty index of Section 5; uncertifiable levels never enter the ladder. **Solve:** the same solver plays perfectly, producing the reference line every model is measured against — it is ranked on the leaderboard as WYRM-BFS, visibly, as the yardstick rather than a competitor. **Play:** each LLM receives a frozen text projection of the board (the grid, worm position and length, apples remaining, keys held, turn budget) plus the complete rule book, and acts one turn at a time; invalid actions are rejected with a machine reason — free information, as in the real game — and every thought, latency and retry is recorded. **Check:** an independent deterministic checker re-executes the recorded action stream on a fresh engine and signs the run CERTIFIED or INVALID; models never grade themselves, and a mismatch voids the artifact. **Rate:** pooled Bradley-Terry MLE over all shared-level battles at once (the Chatbot Arena / Kaggle Game Arena estimator), with 400-round deterministic bootstrap 95% confidence intervals and anchor shrinkage so thin battle matrices stay finite; a solved run beats an unsolved one, efficiency breaks ties, failures draw for half a win each. **Commit & Pages:** run JSON is committed to the repository and a static site re-renders from it, so the leaderboard is a pure function of versioned data.

5 The Difficulty Index (v4)

Difficulty is measured, not asserted. The index replays the certified optimal solution through the real engine and records evidence along eight dimensions (Table 2), each saturating smoothly to 100. Depth — the minimal-turn length of the optimal line — leads the weighting: hardness is first and foremost how long, exact and non-greedy the best plan must be. The composite blends a weighted mean with the strongest dimension and maps to tiers TUTORIAL through NIGHTMARE. Two honesty properties matter more than the formula: a level containing no ice can never score on simulation, and no spikes means no threat score — the index refuses to reward mechanics that are absent. In the current 38-level ladder the mean difficulty is 5.3; the v4 rebalance (depth-led, egg custody added) is deliberately stricter than v3 — no score is grandfathered.

dimension	measured from	meaning
depth	plan length of the optimal line (par)	the headline: minimal turns required
search	BFS states explored (log scale)	combinatorial width of the space
irreversibility	permanent world mutations on the line	digs, burns, unlocks, splits
simulation	forced ice glides to trace	must-look-ahead move sequences
threat	legal moves that would kill (spikes or cracked eggs)	the price of error
choreography	sub-goal ordering load	apples/keys/plates interdependence
custody	egg-care pressure on the winning line	turns carrying + one-wrong-move points
budget	turn-limit pressure	par relative to the limit

Table 2 — The eight dimensions of the v4 difficulty index, each 0-100.

6 The Ladder and Tier V

The ladder is a hand-authored progression of 38 levels whose difficulty curve is enforced, not hoped for: every level carries its computed certificate, and the tiers partition the curve (Table 3). Tier V — the Fifth Circle — was engineered from the primary sources of Section 2: Snakebird's single-wrong-press unwinnability became our threat dimension; Stephen's Sausage Roll's irreversible chains became bridges, dirt and splits that cannot be taken back; the Sokoban proof's corridor gadgets became valve mazes; WOIM's body economy became levels where the certified optimal line requires the worm to sever itself mid-route so its own remains act as slide-stoppers for the next ice glide (Absolute Zero, The Rubicon). The top of the ladder is summarised in Table 4.

tier	levels	character
TUTORIAL	5	rules introduced one at a time
CORE	9	one mechanic under pressure
ADVANCED	12	mechanic pairs, mild irreversibility
GAUNTLET	10	long choreography, real threats
ABYSS	2	deep plans, simulation load, splits

Table 3 — Tier structure of the certified ladder (computed, not declared).

level	tier	difficulty	par/budget	states
The Rubicon (31-the-rubicon)	ABYSS	8.4	38/40	23,992
The Hatchery (38-the-hatchery)	ABYSS	8.0	40/42	185,289
Glass Foundry (22-glass-foundry)	GAUNTLET	7.6	18/20	261,073
Absolute Zero (29-absolute-zero)	GAUNTLET	7.6	20/21	4,287
Cathedral of Lead (24-cathedral-of-lead)	GAUNTLET	7.4	62/65	107,545
The Reliquary (27-the-reliquary)	GAUNTLET	7.4	30/32	44,069

Table 4 — The summit of the ladder: the six highest certified difficulties.

7 Community Governance: the Submission Framework

Anyone can add a level, and no human can quietly break the benchmark. The Level Forge — a browser editor that ships the exact engine, exhaustive validator and difficulty index the ladder uses — lets a contributor paint, certify and playtest a level, then submit it as a GitHub pull request with a contributor card (name, GitHub handle, design intent, tags) embedded in the level file. From there a two-sided review framework takes over.

wyrm-bot, the machine half. On every PR touching the community ladder, a GitHub Actions bot re-certifies each changed level (schema, structure, size and budget policy, slug collisions, exhaustive solvability, certificate honesty) and re-validates the entire existing community ladder so regressions fail too. It posts a structured review card to the PR — contributor, intent, par, difficulty, tier, state count, mechanics on the optimal line, exact failures with fix hints — and labels the PR: bot:validated with bot:awaiting-review, or bot:needs-revision, in which case the submission is returned to the contributor with everything needed to fix it; pushing a fix restarts certification automatically.

Maintainers, the human half. What a bot cannot judge — design quality, fairness, originality, difficulty placement, honest credit — is decided by maintainers listed in the repository's MAINTAINERS file, via slash commands on the PR: /wyrm approve (label bot:approved, auto-merge engages), /wyrm request-changes with a reason (returned with notes), /wyrm revalidate (re-run certification), /wyrm help. On merge the bot posts the contributor a celebration card, the export pipeline picks the level up, and it goes live on the public ladder with the contributor's name on its certificate — then every future benchmark run, every leaderboard refresh, plays it. The whole lifecycle is public record on the site's community board, live from the GitHub API.

8 Model Plug-in Architecture

The arena speaks OpenAI-style chat over pluggable adapters — fourteen provider kinds are built in, including OpenAI, Anthropic, Google, Groq, DeepSeek, Mistral, OpenRouter, xAI, Together, Ollama and LM Studio, plus a custom kind for any OpenAI-compatible endpoint — and wraps them in a production reliability layer: rate limiter (RPM/TPM/concurrency), context fitter, classified retries, circuit breaker and fallback chains, all recorded in a per-run reliability ledger. Any model not reachable this way can still enter over the Model Context Protocol [6]: a zero-dependency MCP bridge exposes the game as tools so MCP-capable clients play without custom code.

Entry is deliberately free. Our verified provider research (November 2026) found ten providers with genuine free tiers and no credit card — Google AI Studio (the most generous: Gemini Flash family, up to ~1,500 requests/day, 1M-token context), Groq, Cerebras, OpenRouter's :free pool, Mistral's Experiment tier (~1B tokens/month), Cloudflare Workers AI, NVIDIA NIM, Cohere's trial keys, SiliconFlow, AnyAPI, and a local Ollama — plus one retired service (GitHub Models, shut down 30 July 2026) and several that require payment details. The full, source-linked table ships on the site's Providers page and in the repository's FREE_PROVIDERS research doc. The benchmark's configuration carries ready-made entries for the free tiers, with rate limits pre-set to live inside them.

9 The Archive

The archive currently holds 38 certified levels and 43 archived runs over 2 enabled model configurations. The reference line — WYRM-BFS, the exhaustive solver — has played the entire ladder at par, certifying every level's solvability claim publicly and providing the baseline against which LLM efficiency is computed (efficiency = par moves / model moves on solved runs). Language-model runs are executed locally by whoever operates the arena and committed as JSON; the site re-renders from the commit. We publish the reference line's replays in full — every step, every state — as worked solutions, the way a textbook publishes answers.

An honesty rule governs the leaderboard: a configured model with zero archived runs is listed as awaiting first run and never shows a default rating; the solver row is ranked visibly but flagged as the reference line, not a competitor. Every model entry links to its independent Artificial Analysis [7] page so arena results can be cross-read against third-party intelligence, speed and price measurements.

10 Reproducibility

Everything is a pure function of committed artifacts. The engine, validator, difficulty index and Elo are single-source: the same TypeScript modules run in CI, in the local benchmark runner, and (compiled to a browser bundle) inside the Level Forge — so a certificate earned in a contributor's browser tab is bit-identical to the one CI computes. The site is static: no backend, no database, no server-side computation; the leaderboard, level atlas, replay viewer (with the model's recorded thoughts alongside the board) and community board are all rendered from committed JSON and the public GitHub API. Rebuilding the entire published artifact from a clean clone is one command; regenerating this paper is another.

11 Limitations and Future Work

Exhaustive certification bounds board size: levels must stay within ~25 million BFS states, which caps how large a single trial can be, though not how deep its plan must be. The text projection is lossy by design — that is the point — but vision models could optionally receive rendered boards, and the vocabulary could grow (eggs, teleports, time manipulation all exist in the source games). The v2 rating stack — pooled Bradley-Terry with bootstrap confidence intervals, anchor shrinkage, capability profiles, progress scores and the mirror-variant registry — was built for the thin-matrix regime, but with very few models the intervals remain wide; battle density is the honest fix. The community framework currently trusts one maintainer list; a CODEOWNERS-based review quorum is a configuration change away. Finally, the ladder measures planning under certainty — the games we drew from have no hidden information, and neither do we; imperfect-information variants remain future work.

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Every reference above was opened and verified on 6 November 2026. This paper is a living artifact: rebuild it with `python3 scripts/build-paper.py`*. Arena, levels, replays, community board and provider research: https://namandhakad712.github.io/WRYM/*